

1. Agrometeorology is a/an:

- a. Pure science
- b. Inter-disciplinary science
- c. Social science
- d. Chemical science

Answer: b

2. The foundation of Agrometeorology is:

- a. Biology of plants
- b. Chemistry of water
- c. Physics of air and soil
- d. Physics of plants

Answer: c

3. Agrometeorology links together:

- a. Soil and water
- b. Physical environment and biological responses
- c. Crops and livestock only
- d. Weather and economics

Answer: b

4. Agrometeorology evolved mainly to understand:

- a. Industrial processes
- b. Effects of meteorological conditions on agriculture
- c. Mining operations
- d. Economic conditions

Answer: b

5. Which is NOT listed as a major disastrous weather phenomenon in the slides?

- a. Cyclones
- b. Floods
- c. Droughts
- d. Earthquakes

Answer: d

6. Crop losses can occur due to aberrations in the:

- a. Physical structure of soil
- b. Temporal march of weather parameters
- c. Chemical composition of crops
- d. Biological cycle of animals

Answer: b

7. Agrometeorology helps improve agricultural planning through:

- a. Accurate weather forecasting
- b. Predicting crop yields and quality
- c. Estimating climatic hazards
- d. All of the above

Answer: d

8. An agro-meteorologist helps farmers make efficient use of their:

- a. Financial environment
- b. Physical environment
- c. Political environment
- d. Social environment

Answer: b

9. Agricultural meteorology studies how weather and climate influence:

- a. Crops only
- b. Livestock only
- c. Crops, livestock, soil, water and farm management
- d. Soil only

Answer: c

10. Which is included in the scope of Agrometeorology?

- a. Crop production
- b. Pest and disease forecasting
- c. Climate-smart agriculture
- d. All of the above

Answer: d

11. Which is an aim of Agrometeorology?

- a. Improve crop yield
- b. Save water
- c. Reduce losses from extreme weather
- d. All of the above

Answer: d

12. Which of the following is a major weather element?

- a. Solar radiation
- b. Temperature
- c. Rainfall
- d. All of the above

Answer: d

13. The major weather elements control:

- a. Germination
- b. Growth
- c. Flowering and yield
- d. All of the above

Answer: d

14. Crop microclimate is the climate:

- a. Above the atmosphere
- b. Within and around the crop canopy
- c. Only inside the soil
- d. Over the entire region

Answer: b

15. Crop microclimate is influenced by:

- a. Plant height
- b. Plant spacing
- c. Soil moisture and management
- d. All of the above

Answer: d

16. Which of the following is related to radiation and energy balance?

- a. Direct solar radiation
- b. Diffuse solar radiation
- c. Reflection (albedo)
- d. All of the above

Answer: d

17. Net radiation drives:

- a. Infiltration only
- b. Evaporation and plant growth
- c. Runoff only
- d. Groundwater recharge only

Answer: b

18. Heat moves through soil by:

- a. Conduction
- b. Transpiration
- c. Evaporation
- d. Runoff

Answer: a

19. Soil temperature changes:

- a. Daily only
- b. Seasonally only
- c. Daily and seasonally
- d. Never changes

Answer: c

20. Soil temperature affects:

- a. Seed germination and root growth
- b. Cloud formation only
- c. Wind speed
- d. Rainfall only

Answer: a

21. Which is a component of the hydrological cycle mentioned in the slides?

- a. Rainfall
- b. Infiltration
- c. Runoff
- d. All of the above

Answer: d

22. Soil water budget helps in:

- a. Weather forecasting only
- b. Irrigation scheduling

- c. Pest identification only
- d. Wind measurement

Answer: b

23. Which of the following is a weather hazard mentioned in the slides?

- a. Drought
- b. Frost
- c. Hail
- d. All of the above

Answer: d

24. Climate change increases:

- a. Temperature
- b. Rainfall variability
- c. Extreme events
- d. All of the above

Answer: d

25. Which is an adaptation measure to climate change mentioned in the slides?

- a. Resilient crops
- b. Efficient irrigation
- c. Weather advisories
- d. All of the above

Answer: d

26. LST stands for:

- a. Local Soil Temperature
- b. Land Surface Temperature
- c. Long-term Soil Temperature
- d. Land Seasonal Temperature

Answer: b

27. NDVI is associated with:

- a. Crop health
- b. Wind speed
- c. Rainfall measurement only
- d. Groundwater recharge

Answer: a

28. ENSO includes:

- a. El Niño and La Niña
- b. IOD and MJO
- c. SPI and SMI
- d. GDD and LST

Answer: a

29. Nowcasting covers:

- a. 0–6 hours
- b. 1–3 days
- c. 4–10 days
- d. 2–4 weeks

Answer: a

30. Which forecast range is associated with sowing and harvesting?

- a. Nowcasting
- b. Short-range
- c. Medium-range
- d. Extended-range

Answer: c

Meteorology

Atmosphere + Physics + Geography

Weather

Given place + given time

Climate

Long-term

Agrometeorology

Applied meteorology + agriculture

Meteorology vs Agrometeorology

Weather forecasting vs crop production/agro-advisory

Atmosphere

90% mass → 15 km

Mass → 5.15×10^{18} kg

Layers

Troposphere ↓ → Stratosphere ↑ → Mesosphere ↓

Heights

10–15 → 45–55 → 80–90 km

Lapse rate

Dry = 10°C/km

Moist = 6°C/km

Climate normal

30 years

Altitude

6.5°C/km decrease

Rainfall zones

<500 Arid → 500–750 Semi-arid → 750–1000 Sub-arid → >1000 Humid

Soil

Solid + Liquid + Gas

Texture

Sand > Silt > Clay

Settling

Sand 1 min → Silt 3–4 h → Clay 3–7 days

Soil texture triangle

12 classes

Thermal properties

Conductivity → Heat conduction

Heat capacity → Energy for temperature rise

Diffusivity → Rate of temperature transmission

Climate scales

Micro < Meso < Macro

Microclimate

Weather experienced by crop

Microclimate = link

Weather ↔ Crop

Moisture Index

$Im = 100(P-PE)/PE$

India

15 Agro-Climatic Zones

ACZ based on

Temperature + Rainfall + Soil + Crop suitability

1. Meteorology is a branch of science dealing with:

- a. Soil**
- b. Atmosphere**
- c. Crops**
- d. Water**

Answer: b

2. Meteorology is a combination of:

- a. Biology and chemistry**
- b. Physics and geography**

- c. Chemistry and geography
- d. Physics and biology

Answer: b

3. Weather refers to the:

- a. Long-term atmospheric condition of a region
- b. Physical state of atmosphere at a given place and time
- c. Average rainfall of a region
- d. Average temperature over 30 years

Answer: b

4. Climate refers to:

- a. Atmospheric condition at a particular instant
- b. Short-term weather variation
- c. Long-term regime of atmospheric variables of a place/area
- d. Daily temperature variation

Answer: c

5. Agricultural meteorology is primarily a branch of:

- a. Pure meteorology
- b. Applied meteorology
- c. Physical chemistry
- d. Hydrology

Answer: b

6. Agricultural meteorology deals with the relationship between:

- a. Weather/climatic conditions and agricultural production
- b. Soil and geology only
- c. Animals and soil only
- d. Water and atmosphere only

Answer: a

7. Agricultural meteorology is described as a:

- a. Physical science
- b. Biophysical science
- c. Chemical science
- d. Geological science

Answer: b

8. Meteorology primarily aims at:

- a. Improving crop quality
- b. Weather forecasting
- c. Crop planning
- d. Soil management

Answer: b

9. Agricultural meteorology primarily aims at improving:

- a. Weather forecasting only
- b. Quantity and quality of crop production
- c. Soil colour
- d. Atmospheric pressure

Answer: b

10. Weather and climate are considered _____ in agricultural planning.

- a. Secondary factors
- b. Basic inputs/resources
- c. Unimportant factors
- d. Biological factors

Answer: b

11. Which of the following is NOT listed as a farm operation affected by weather?

- a. Ploughing
- b. Irrigation
- c. Harvesting
- d. Mineral exploration

Answer: d

12. Which climatic factors are important for determining crop growing season?

- a. Solar radiation, air temperature, precipitation, wind and humidity
- b. Only temperature and rainfall
- c. Soil pH and texture only
- d. Wind and soil colour only

Answer: a

13. Crop planning for stability in production aims to:

- a. Increase climatic risk
- b. Reduce risk of crop failure due to climate
- c. Eliminate soil moisture
- d. Increase weather adversity

Answer: b

14. Which of the following is considered while selecting suitable crops/cropping patterns?

- a. Crop water requirement
- b. Effective rainfall
- c. Available soil moisture
- d. All of the above

Answer: d

15. Soil moisture stress and drought can be diagnosed using:

- a. Climatic water balance method
- b. Soil texture triangle
- c. Weather forecasting only
- d. Soil colour method

Answer: a

16. Which of the following is NOT listed as a climatic extremity?

- a. Frost
- b. Floods
- c. Droughts
- d. Earthquake

Answer: d

17. When rain is forecast, according to the slide, farmers should:

- a. Apply irrigation
- b. Avoid irrigation
- c. Apply more fertilizer
- d. Apply anti-transpirant

Answer: b

18. When frost is forecast, the slide recommends:

- a. Avoiding irrigation
- b. Applying irrigation

- c. Avoiding harvesting
- d. Applying fertilizer

Answer: b

19. When rain is forecast, fertilizer and chemical sprays should be:

- a. Increased
- b. Avoided
- c. Doubled
- d. Applied immediately

Answer: b

20. Shelter belts are mentioned for mitigation against:

- a. Floods and droughts
- b. Cold waves and heat waves
- c. Rainfall and humidity
- d. Pests and diseases

Answer: b

21. The lower atmosphere mentioned in the slide extends up to approximately:

- a. 5 km
- b. 10 km
- c. 20 km
- d. 50 km

Answer: c

22. Approximately what percentage of atmospheric mass is within 15 km of Earth's surface?

- a. 50%
- b. 75%
- c. 90%
- d. 100%

Answer: c

23. The approximate mass of Earth's atmosphere is:

- a. 5.15×10^{15} kg
- b. 5.15×10^{18} kg

- c. 5.15×10^{21} kg
- d. 5.15×10^{12} kg

Answer: b

24. The lowest layer of the atmosphere is:

- a. Stratosphere
- b. Mesosphere
- c. Troposphere
- d. Thermosphere

Answer: c

25. In the troposphere, temperature generally:

- a. Increases with height
- b. Decreases with height
- c. Remains constant
- d. First increases and then remains constant

Answer: b

26. The tropopause is approximately at:

- a. 1–5 km
- b. 10–15 km
- c. 45–55 km
- d. 80–90 km

Answer: b

27. In the stratosphere, temperature:

- a. Decreases with altitude
- b. Increases with altitude
- c. Remains constant
- d. Becomes zero

Answer: b

28. The mesopause is:

- a. The hottest point in the atmosphere
- b. The coldest point in the atmosphere
- c. The lowest atmospheric layer
- d. The highest pressure region

Answer: b

29. The dry-adiabatic lapse rate given in the slide is:

- a. $2^{\circ}\text{C km}^{-1}$
- b. $6^{\circ}\text{C km}^{-1}$
- c. $10^{\circ}\text{C km}^{-1}$
- d. $15^{\circ}\text{C km}^{-1}$

Answer: c

30. The moist-adiabatic lapse rate given in the slide is:

- a. $4^{\circ}\text{C km}^{-1}$
- b. $6^{\circ}\text{C km}^{-1}$
- c. $8^{\circ}\text{C km}^{-1}$
- d. $10^{\circ}\text{C km}^{-1}$

Answer: b

31. Climatic normals are generally worked out for:

- a. 5 years
- b. 10 years
- c. 20 years
- d. 30 years

Answer: d

32. The general temperature decrease with altitude given in the slide is:

- a. 2.5°C/km
- b. 4.5°C/km
- c. 6.5°C/km
- d. 10°C/km

Answer: c

33. Rainfall of less than 500 mm corresponds to:

- a. Humid
- b. Sub-arid
- c. Semi-arid
- d. Arid

Answer: d

34. Which statement about soil texture is correct?

- a. Clay particles are largest
- b. Silt particles are smallest

- c. Sand particles are largest
- d. Sand, silt and clay are equal in size

Answer: c

35. India has how many agro-climatic zones according to the slide?

- a. 10
- b. 12
- c. 15
- d. 20

Answer: c



LECTURE 2 — ULTRA-FAST MEMORY SHEET

These are the facts I would memorize first for MCQs:

Solar radiation

- Insolation → electromagnetic energy from Sun
- Solar constant → $\sim 1373 \text{ W m}^{-2}$
- Atmosphere can reduce incoming radiation by 15–80%
- UV → stress/damage
- Visible → photosynthesis
- IR → thermal energy

PAR

- 400–700 nm
- Night → 0
- Summer midday → 2000–3000
- Plants → PAR
- Humans → lumen/lux

Chlorophyll

- Chlorophyll b → blue
- Chlorophyll a → red
- Photosynthesis peaks → 400–460 nm and 580–700 nm

PPFD

- Measures photons reaching plant

- Unit $\rightarrow \mu\text{mol m}^{-2} \text{s}^{-1}$
- Seedlings $\rightarrow$ lower PPFD
- Flowering $\rightarrow$ higher PPFD

Radiation

- Absorption + reflection + transmission
- Black body $\rightarrow$ perfect absorber + emitter
- Albedo $\rightarrow$ reflected radiation
- Fresh snow $\rightarrow$ very high albedo
- Dark wet surface $\rightarrow$ low albedo

Radiation laws

- Planck $\rightarrow E = h\nu$
- Kirchhoff $\rightarrow$ good absorber = good emitter
- Stefan-Boltzmann $\rightarrow E \propto T^4$
- Wien $\rightarrow \lambda_{\text{max}} \propto 1/T$
- Frequency $\uparrow \rightarrow$ wavelength $\downarrow$
- Frequency $\uparrow \rightarrow$ energy $\uparrow$

Radiation budget

$$Q^* = S_{\downarrow} - S_{\uparrow} + L_{\downarrow} - L_{\uparrow} \quad Q^* = S_{\downarrow} - S_{\uparrow} + L_{\downarrow} - L_{\uparrow}$$

- DSR $\rightarrow$ incoming solar
- USR $\rightarrow$ reflected solar
- DLR $\rightarrow$ incoming longwave
- ULR $\rightarrow$ outgoing longwave

Heat balance

$$R_n = H + LE + G + P + MR_n = H + LE + G + P + M$$

Practically:

$$R_n \approx H + LE + G \quad R_n \approx H + LE + G$$

- H $\rightarrow$ sensible heat
- LE $\rightarrow$ latent heat
- G $\rightarrow$ soil/ground heat

Scattering

- Rayleigh $\rightarrow$ small particles $\rightarrow$ blue sky
- Mie $\rightarrow$ larger particles
- Sunset/sunrise red $\rightarrow$ longer atmospheric path + greater scattering of short wavelengths

Light

- Visible range in slide → 0.39–0.76 μm
- 1 foot-candle → 10.764 lux
- Lux meter → measures light intensity
- Light compensation point → photosynthesis = respiration

Plants

- Sciophytes → shade-loving → ~20% sunlight saturation
- Heliophytes → sun-loving
- Sciophyte examples → maize, potato, tomato, herbs
- Heliophyte examples → sunflower, banyan

Photoperiodism

- Long day → >12 h
- Short day → <12 h
- Day neutral → unaffected by photoperiod
- Long-day examples → potato, sugar beet, wheat, barley
- Short-day → rice, sorghum, cotton, sunflower
- Day-neutral → tomato, maize

Plant responses

- Photomorphogenesis → plant development response to light
- Phototropism → movement/growth response to light
- Positive → towards light
- Negative → away from light

Air temperature

Factors:

Latitude + Altitude + Land/water + Ocean currents + Winds + Clouds + Mountains + Surface + Relief + Convection

🔥 Cloudy day → cooler

🔥 Cloudy night → warmer

🔥 Mountain top → colder than valley

Lecture 2 — 35 Most Important MCQs

These cover the maximum important concepts, definitions, formulas, numerical values, classifications, and likely MCQ traps from Lecture 2. Only slide content is used.

1. Insolation refers to:

- a. Heat emitted by Earth
- b. Electromagnetic energy radiated by the Sun into space
- c. Heat stored in soil
- d. Energy emitted by clouds

Answer: b

2. The approximate value of the solar constant is:

- a. 973 W m^{-2}
- b. 1173 W m^{-2}
- c. 1373 W m^{-2}
- d. 1573 W m^{-2}

Answer: c

3. Which of the following affects insolation?

- a. Cloud cover
- b. Albedo
- c. Length of day and night
- d. All of the above

Answer: d

4. The solar constant mainly depends on:

- a. Soil moisture and temperature
- b. Energy output of Sun and Earth-Sun distance
- c. Cloudiness and wind speed
- d. Latitude and altitude

Answer: b

5. Reflection, scattering and absorption by the atmosphere can reduce solar radiation reaching Earth's surface by approximately:

- a. 5–10%
- b. 15–80%
- c. 80–95%
- d. 95–100%

Answer: b

6. Which spectral region is primarily responsible for photosynthesis?

- a. Ultraviolet
- b. Visible
- c. Infrared
- d. Microwave

Answer: b

7. Infrared radiation is mainly associated with:

- a. Photosynthesis
- b. Thermal energy exchange
- c. DNA formation
- d. Seed germination only

Answer: b

8. Excessive ultraviolet radiation can:

- a. Increase plant growth indefinitely
- b. Damage plant tissues and cellular components
- c. Stop transpiration completely
- d. Increase PAR

Answer: b

9. Photosynthetically Active Radiation (PAR) lies approximately in the wavelength range:

- a. 200–400 nm
- b. 300–500 nm
- c. 400–700 nm
- d. 700–1000 nm

Answer: c

10. PAR is greatest generally during:

- a. Winter midnight
- b. Summer midday
- c. Winter evening
- d. Sunset

Answer: b

11. Which of the following reduces PAR reaching plants?

- a. Cloud cover
- b. Tree shading
- c. Buildings
- d. All of the above

Answer: d

12. At night, the PAR value is:

- a. 100
- b. 500
- c. 1000
- d. Zero

Answer: d

13. During summer midday, PAR often reaches approximately:

- a. 100–500
- b. 500–1000
- c. 2000–3000
- d. 4000–5000

Answer: c

14. Chlorophyll b is most active with:

- a. Red light
- b. Blue light
- c. Infrared radiation
- d. Ultraviolet radiation

Answer: b

15. Chlorophyll a is most active with:

- a. Blue light
- b. Green light
- c. Red light
- d. UV light

Answer: c

16. Photosynthesis peaks approximately at:

- a. 200–300 nm and 700–800 nm
- b. 400–460 nm and 580–700 nm
- c. 500–550 nm only
- d. 700–900 nm only

Answer: b

17. PPFD stands for:

- a. Photosynthetic Plant Flux Density
- b. Photosynthetic Photon Flux Density
- c. Plant Photosynthetic Frequency Distribution
- d. Photosynthetic Power Flux Distribution

Answer: b

18. PPFD is expressed in:

- a. W m^{-2}
- b. lux
- c. $\mu\text{mol m}^{-2} \text{s}^{-1}$
- d. J m^{-3}

Answer: c

19. PPFD represents:

- a. Number of photons reaching a defined surface per second within the PAR range
- b. Total heat emitted by plants
- c. Amount of infrared radiation emitted
- d. Amount of light visible only to humans

Answer: a

20. Which statement about PPFD is correct?

- a. Seedlings generally require higher PPFD than flowering plants
- b. Seedlings generally require lower PPFD than mature plants
- c. Flowering requires lower PPFD than seedlings
- d. PPFD is independent of plant variety

Answer: b

21. The ratio of reflected radiation to incident radiation is called:

- a. Absorptivity
- b. Emissivity
- c. Albedo
- d. Transmissivity

Answer: c

22. A black body has:

- a. Absorptivity = 0 and emissivity = 0
- b. Absorptivity = 1 and emissivity = 1
- c. Absorptivity = 0 and emissivity = 1
- d. Absorptivity = 1 and emissivity = 0

Answer: b

23. Which surface has the highest albedo among the values given in the slide?

- a. Open ocean
- b. Grass
- c. Fresh snow
- d. Dark wet soil

Answer: c

24. Albedo of fresh snow is approximately:

- a. 0.05
- b. 0.20
- c. 0.40
- d. 0.85–0.90

Answer: d

25. According to Kirchhoff's law:

- a. Good absorber is a poor emitter
- b. Good absorber is also a good emitter
- c. Radiation is independent of wavelength
- d. Emission is proportional to temperature

Answer: b

26. According to Stefan-Boltzmann law, emitted radiation is proportional to:

- a. T
- b. T^2
- c. T^3
- d. T^4

Answer: d

27. The Stefan-Boltzmann constant is approximately:

- a. $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
- b. $6.62 \times 10^{-34} \text{ J s}$
- c. $2.897 \times 10^3 \mu\text{m K}$
- d. 1373 W m^{-2}

Answer: a

28. Wien's displacement law states that maximum emission wavelength is:

- a. Directly proportional to absolute temperature
- b. Inversely proportional to absolute temperature
- c. Independent of temperature
- d. Proportional to temperature²

Answer: b

29. According to Planck's law, energy of a quantum is:

- a. $E = mc^2$
- b. $E = h\nu$
- c. $E = \sigma T^4$
- d. $E = \lambda T$

Answer: b

30. If frequency increases, wavelength:

- a. Increases
- b. Decreases
- c. Remains constant
- d. Becomes zero

Answer: b

31. Rayleigh scattering is primarily responsible for:

- a. Red colour of sunset only
- b. Blue colour of the sky

- c. Green colour of plants
- d. Infrared emission

Answer: b

32. The light compensation point is the light intensity at which:

- a. Photosynthesis is maximum
- b. Respiration is zero
- c. Photosynthesis equals respiration
- d. Transpiration equals evaporation

Answer: c

33. Sciophytes are:

- a. Sun-loving plants
- b. Shade-loving plants
- c. Water-loving plants
- d. Desert plants

Answer: b

34. Which of the following is a short-day plant according to the slide?

- a. Wheat
- b. Barley
- c. Rice
- d. Potato

Answer: c

35. Photoperiodism refers to the response of plants to:

- a. Temperature
- b. Relative length of day and night
- c. Soil moisture
- d. Wind direction

Answer: b

